HIGHER DIRECT IMAGES OF A QUASI-COMPACT QUASI-SEPARATED MORPHISM COMMUTE WITH FILTERED COLIMITS OF ABELIAN SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The small etale topos of an algebraic space as the topos of the site of etale maps. Properties of Spaces, label spaces-properties-lemma-functoriality-etale-site (invoked at proof line 716). It identifies the morphism of topoi f_{small} with the one induced by the morphism of sites $f_{spaces,\acute{e}tale}$ attached to the continuous functor $V \mapsto X \times_Y V$, so that the general criterion for morphisms of sites becomes available.
- Colimits of higher direct images along a morphism of sites. Cohomology on Sites, label sites-cohomology-lemma-higher-direct-image-colimit (invoked at proof line 719). It reduces the claim to the corresponding statement on the members of a covering of every object of the target site, which here may be taken affine.
- Cohomology of a quasi-compact quasi-separated algebraic space commutes with filtered colimits. Same chapter spaces-cohomology.tex, label lemma-colimits (invoked at proof line 726). It settles the remaining statement for the quasi-compact quasi-separated space $X \times_Y V$.

Source reference. Original lemma 073E: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 667–667. Navigation pointer for \ref{lemma-colimits} used inside the counted span (line 726); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0H7B: sources/stacks-chow/tags.txt, source lines 21451–21451. Navigation pointer for \ref{sites-cohomology-lemma-higher-direct-image-colimit} used inside the counted span (line 719); the argument that consumes it is retained in the span.

Source reference. Original lemma 03G2: sources/stacks-chow/tags.txt, source lines 4513–4513. Navigation pointer for \ref{spaces-properties-lemma-functoriality-etale-site} used inside the counted span (line 717); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of algebraic spaces over S. Let $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ be a filtered colimit of abelian sheaves on $X_{étale}$. Then for any $p \geq 0$ we have*

$$R^p f_* \mathcal{F} = \operatorname{colim} R^p f_* \mathcal{F}_i.$$

Proof. We will use that the morphism of topoi $f_{small} : X_{small} \to Y_{small}$ comes from the morphism of sites $f_{spaces,étale} : X_{spaces,étale} \to Y_{spaces,étale}$ corresponding to the continuous functor $V \longmapsto X \times_Y V$, see Properties of Spaces, Lemma 03G2. We will apply Cohomology on Sites, Lemma 0H7B to this morphism of sites. Since every object of $Y_{spaces,étale}$ has a covering by affine objects, it suffices to show that for V affine and étale over Y we have $H^p(X \times_Y V, \mathcal{F}) = \operatorname{colim} H^p(X \times_Y V, \mathcal{F}_i)$. Since V is affine, the algebraic space $X \times_Y V$ is quasi-compact and quasi-separated. Hence we can apply Lemma 073E to conclude. □